(1) Let $q_1 = \frac{a-c}{4}$, $q_2 = \frac{a-c}{4}$

Then

$$\Pi_1 = \Pi_2 = \left[a - \left(\frac{a-c}{4} + \frac{a-c}{4}\right)\right]\frac{a-c}{4} - c \cdot \frac{a-c}{4}$$

$$= \frac{(a-c)^2}{8}.$$

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$$\Pi_1 = \left[a - \left(\frac{a-c}{4} + \frac{a-c}{3}\right)\right]\frac{a-c}{4} - c \cdot \frac{a-c}{4}$$

$$= \frac{5}{48}(a-c)^2.$$

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$$= \frac{5}{48}(a-c)^2.$$

Let $q_1 = \frac{(a-c)}{3}$, $q_2 = \frac{a-c}{3}$

Then

$$\Pi_1 = \Pi_2 = \left[a - \left(\frac{a-c}{3} + \frac{a-c}{3}\right)\right]\frac{a-c}{3} - c \cdot \frac{a-c}{3}$$

$$= \frac{(a-c)^2}{9}.$$

So the normal form of the game is

Firm 1 \ Firm 2	$\frac{a-c}{4}$ (L)	$\frac{a-c}{3}$ (H).
$\frac{a-c}{4}$ (L)	$\frac{(a-c)^2}{8}, \frac{(a-c)^2}{8}$	$\frac{5}{48}(a-c)^2, \frac{5}{36}(a-c)^2$
$\frac{a-c}{3}$ (H)	$\frac{5}{36}(a-c)^2, \frac{5}{48}(a-c)^2$	$\frac{(a-c)^2}{9}, \frac{(a-c)^2}{9}$.

Note that $\frac{5}{48}(a-c)^2 < \frac{1}{9}(a-c)^2 < \frac{1}{8}(a-c)^2 < \frac{5}{36}(a-c)^2$

Hence, the dominant strategy for each firm is to produce $q = \frac{a-c}{3}$, which is higher than $q = \frac{a-c}{4}$. ~~as~~ The equilibrium pay-offs for each are $\frac{(a-c)^2}{9}$. However, had they co-operated and produced a lower $q = \frac{a-c}{4}$, they would have each made higher profits equal to $\frac{(a-c)^2}{8}$.

Ques 2 transfer rule : $t(a_2)$

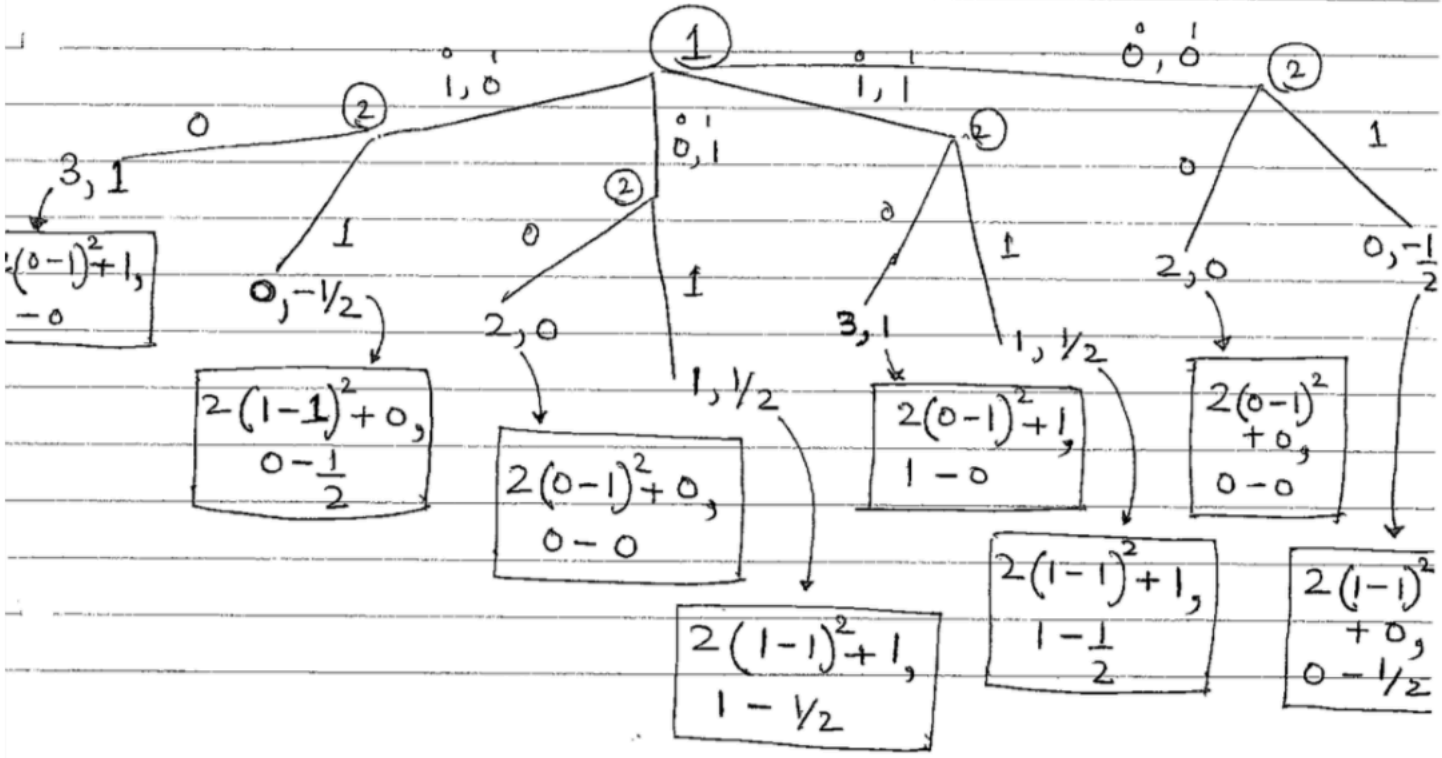
$a_2=0$	$a_2=1$
1 ,	0
0 ,	1
1 ,	1
0 ,	0

Player 2 : payoffs = t − cost
(Max.)

cost: $0\ (a_2=0)$, $\frac{1}{2}\ (a_2=1)$

ie we have,
$t-0$ or $t-\frac{1}{2}$ as the payoffs of player 2.

Player 1 : payoffs = $2(a_2-1)^2+t$
(Min).



(3) Player 1's strategy s_1 can be between 0 or 1.
$s_1 \in [0, 1]$.

Similarly for Pl 2,
$s_2 \in [0, 1]$.

~~A~~ Player i maximises $U_i = s_1 s_2 - \frac{s_i^2}{10}$.

<u>Best Response for Pl 1</u>

$$\max \quad U_1 = s_1 s_2 - \frac{s_1^2}{10}.$$

$$\frac{dU_1}{ds_1} = s_2 - \frac{s_1}{5} = 0$$

$$\Rightarrow \quad s_2 = \frac{s_1}{5}.$$

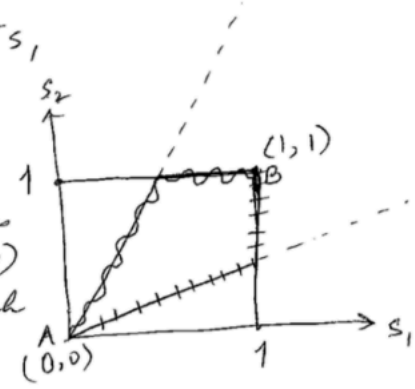
<u>Best Response for Pl 2</u>

$$\text{Max} \quad U_2 = s_1 s_2 - \frac{s_2^2}{10}.$$

$$\frac{dU_2}{ds_2} = s_1 - \frac{s_2}{5} = 0$$

$$\Rightarrow \quad s_2 = 5 s_1$$

Plotting the Best Response Functions,



The 2 Best Response functions intersect at 2 points A(0,0) & B (1,1) so there are 2 pure strategy Nash Equilibria (0,0) and (1,1).

Ans
(4)

For Player 1, $x_1 \in [-1, 1]$

For Player 2, $x_2 \in [-1, 1]$.

Pl 1 maximises $u_1 = \frac{x_1^2}{2} - x_1 x_2$

Pl 2 maximises $u_2 = \frac{x_2^2}{2} + x_1 x_2$.

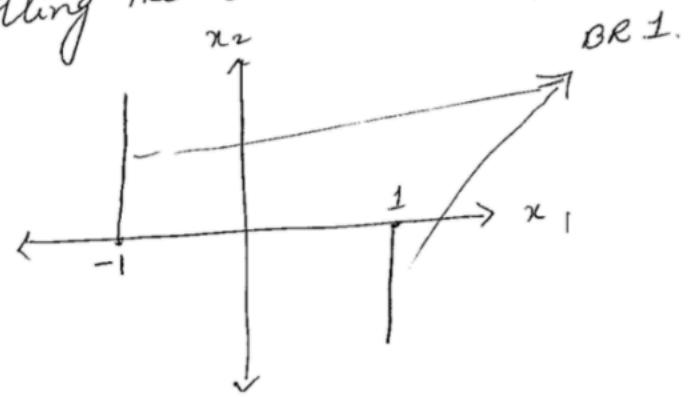
<u>Best Response of Pl 1</u>:

When $x_2 < 0$,
$x_1 = 1$ is the Best Response.

When $x_2 = 0$,
$x_1 = \pm 1$. is the Best Response.

When $x_2 > 0$,
$x_1 = -1$ is the Best Response.

Plotting the Best Response of Pl 1:



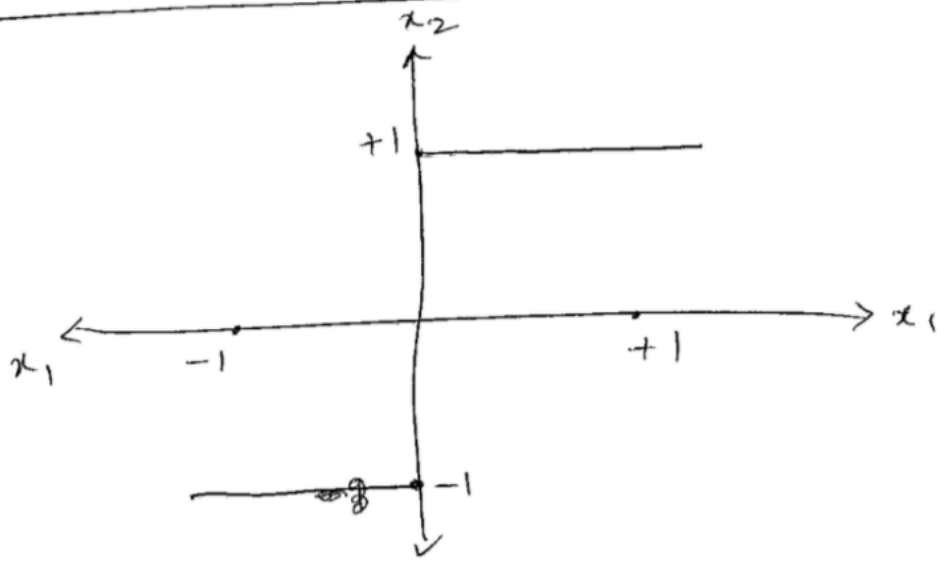
Best Response for Pl 2,

When $x_1 < 0$, $x_2 = -1$ is the Best Response

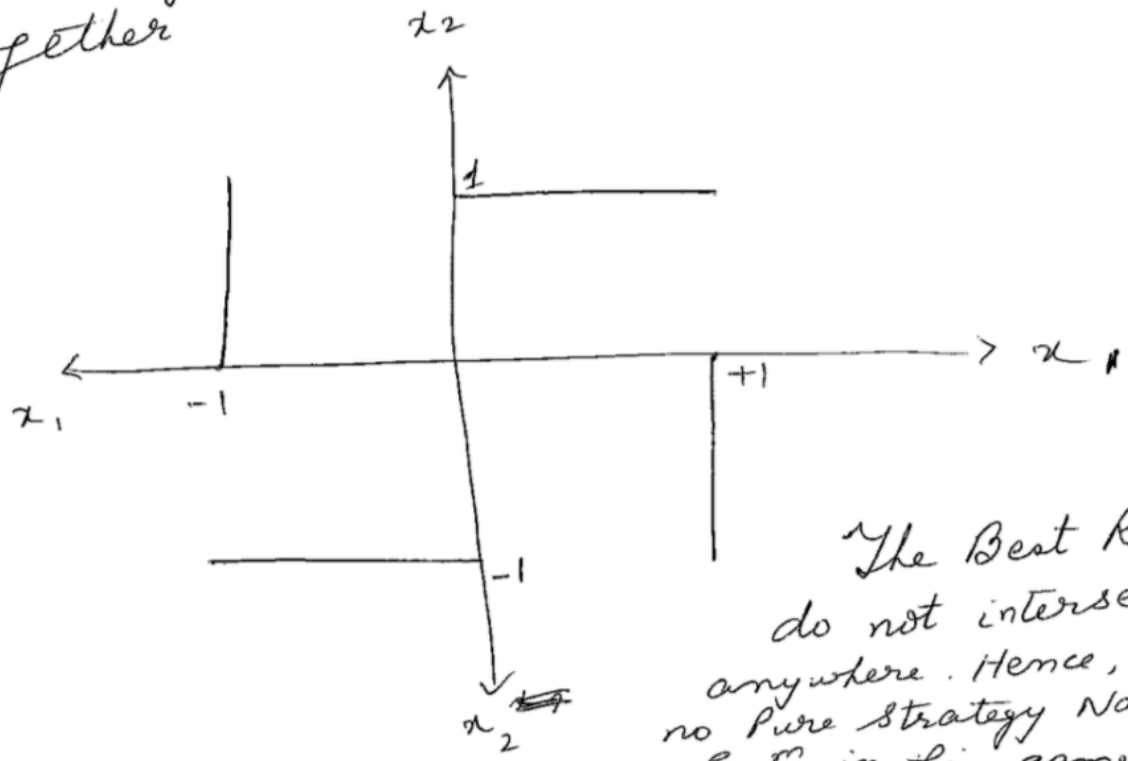
" $x_1 = 0$, $x_2 = \pm 1$ " " " "

" $x_1 > 0$, $x_2 = 1$ " " " "

Best Response of Pl 2.



Plotting both the Best Responses of Pl 1 & Pl 2 together



The Best Responses do not intersect anywhere. Hence, there is no Pure Strategy Nash Eq^m in this game